SWARAJ SANTOSH VITE

B.Tech. - Chemical Engineering

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Junnar, Maharashtra, India - 410504

EDUCATION

MIT Academy of Engineering, Pune — 2021 - 2026

B.Tech. - Chemical Engineering | CGPA: **6.95** / **10**

Guruvarya R.P.Sabnis Vidyamandir,Narayangaon, Junnar — 2021

12th | MSBSHSE | Percentage: **72.33** / **100**

BLOOMINGDALE INTERNATIONAL SCHOOL,NARAYANGAON, Junnar — 2019

10th | CISCE | Percentage: **78** / **100**

INTERNSHIPS

Katraj Dairy | Agriculture / Dairy / Forestry / Fishing — 01 Jul, 2025 - 31 Jul, 2025

Intern

Key Skills:

Chilling | Homogenization | Sterilization Process | Quality Testing and Analysys | Milk Processing | Pasteurization

Completed a one-month summer internship at Katraj Dairy, gaining exposure to milk processing operations like pasteurization, homogenization, quality testing of milk, steam generation process using boiler, and food safety standards, quality control procedures, and basic dairy plant equipment's, where I have observed complete plant operability, also I have observed clarification process, cream separation process. After all the process we get our dairy products that have maximum lifespan with a suitable cost.

PROJECTS

Cellulose Extraction from Waste Banana Stem — 02 Mar, 2026 - 23 May, 2026

Mentor: Prof. Dr. Suraj V. Yadav | **Team Size:** 1

Key Skills:

Chemical Processing Skills | Laboratory Skills | Analytical Skills | Characterization | Agro-waste utilization | Environmental sustainability

Worked on extracting cellulose from banana stem waste as part of a sustainable chemical engineering project. Applied pre-treatment, alkali treatment, bleaching, and purification processes to isolate cellulose. Conducted analysis using characterization techniques to evaluate purity and structural properties. Focused on waste valorization and eco-friendly material development aligned with circular economy principles.
Extracted cellulose from banana stem waste using chemical treatment methods. Gained hands-on experience in biomass processing, purification, and material characterization techniques for sustainable applications.

Smart Water Distribution and Networking in Apartment — 06 Jan, 2025 - 27 Dec, 2025

Mentor: Prof.Dr.Mayurkumar.P.Patil | **Team Size:** 3

Key Skills: Modelling and Simulation | Use of Sensor Technology | Recycle and Reuse of Wastewater | Save the water

Efficient water distribution is a critical challenge in apartment complexes and societies, where water demand is high, and equitable distribution is essential for the well-being of residents. The working system that ensures equitable water distribution in all apartments within a society. The proposed solution leverages smart sensors, real-time data monitoring, and automated systems to optimize water use, detect leakages, and adjust water pressure to reduce wastage in water and provide uniform distribution. The introduction of Internet of Things technology in water distribution networks improves monitoring and control facilities and enables a dynamic response to varying water requirements. This way, water saving methods can be implemented using sustainability as a criterion, and overall efficiency can be increased because these systems can attain optimum performance with minimal effort.

CFD Simulation of Double Pipe Heat Exchanger — 12 Sep, 2022 - 17 Jun, 2023

Mentor: Prof.Dr.Mayurkumar.P.Patil | **Team Size:** 4

Key Skills: Modelling and Simulation | Autocad | Ansys Fluent Software | Computational Fluid Dynamics

Simulation of heat transfer mechanisms in the heat exchangers is very important in different industrial process to recover and minimize the wastage of thermal energy. Ansys Fluent is well-known software for computational fluid dynamics (CFD) which is used to simulate

and analyze heat transfer mechanisms in the heat exchanging system. In the present work, a model of simulation for heat transfer rate observed from the double pipe heat exchanger (DPHE) has been developed using Ansys Fluent 16.2 software.

ASSESSMENTS / CERTIFICATIONS

SWAYAM-NPTEL

Aggregate: 51 / 100
Subjects: Membrane Technology (51 / 100)
A membrane technology course provides a comprehensive understanding of membrane-based separation processes and their applications across various industries. The course covers the fundamentals of membrane science, including membrane materials, their properties, and preparation methods. It delves into the principles of different membrane processes like reverse osmosis, microfiltration, ultrafiltration, and pervaporation, along with their practical applications.

SEMINARS / TRAININGS / WORKSHOPS

Process Safety: HAZOP Studies

05 Apr, 2026 - 05 Apr, 2026

Institute Name: CHEMKLUB INDIA PVT LTD

Key Skills: Hazard and Operability Studies | Process Safety

Unlocking the Power of Aspen Hysys to Optimize & Innovate: Sustainability Model

20 Aug, 2025 - 20 Aug, 2025

Institute Name: MIT Academy of Engineering, Pune

Key Skills: Aspen HYSYS | Simulation | Energy Optimization and Utilization

Gained hands-on experience in process simulation, optimization, and sustainability modeling using Aspen HYSYS®. Learned to design and analyze energy-efficient and environmentally sustainable chemical processes. Successfully completed the workshop and received a certificate of participation/completion.

Aspen HYSYS Refresher Course

21 Mar, 2025 - 21 Mar, 2025

Institute Name: MIT Academy of Engineering, Pune

Key Skills: Aspen HYSYS | Simulation of Distillation Column

The workshop is mainly designed to help participants brush up on their skills in using Aspen HYSYS, a process simulation software. It typically involve a combination of live instruction, hands on exercises, and access to a virtual training lab. It focuses on key areas like fluid packages, heat exchangers, distillation column modelling.

Achieve Sustainability Goals with Aspen Plus Solution

06 Feb, 2025 - 06 Feb, 2025

Institute Name: MIT Academy of Engineering, Pune

Key Skills: Modelling and Simulation | Sustainability | Energy analysys | Aspen Plus Simulation software

This Aspen plus simulation workshop focuses on utilizing the software to achieve sustainability goals in process industries. We learn how to model and optimize processes to reduce energy consumption, minimize waste, lower emissions, and identify green energy sources. The workshop cover topics like carbon capture, energy analysis, and biomass applications, with a strong emphasis on practical case studies.

PERSONAL INTERESTS / HOBBIES

- Cycling, Travelling, Basketball, Football.

IMs

- WhatsApp - https://wa.me/<9987704240>
- WhatsApp - https://wa.me/<8856839709>

PERSONAL DETAILS

Gender: Male
Marital Status: Single
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